

Git and Github for Bio724

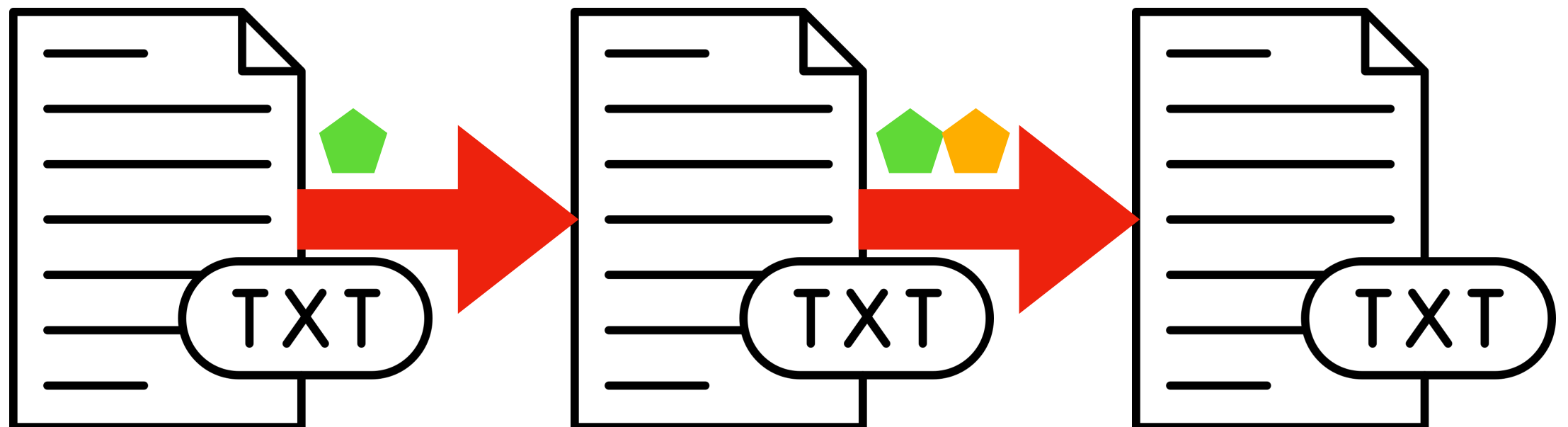
Paul M. Magwene

**Git is a
"distributed version control system"**

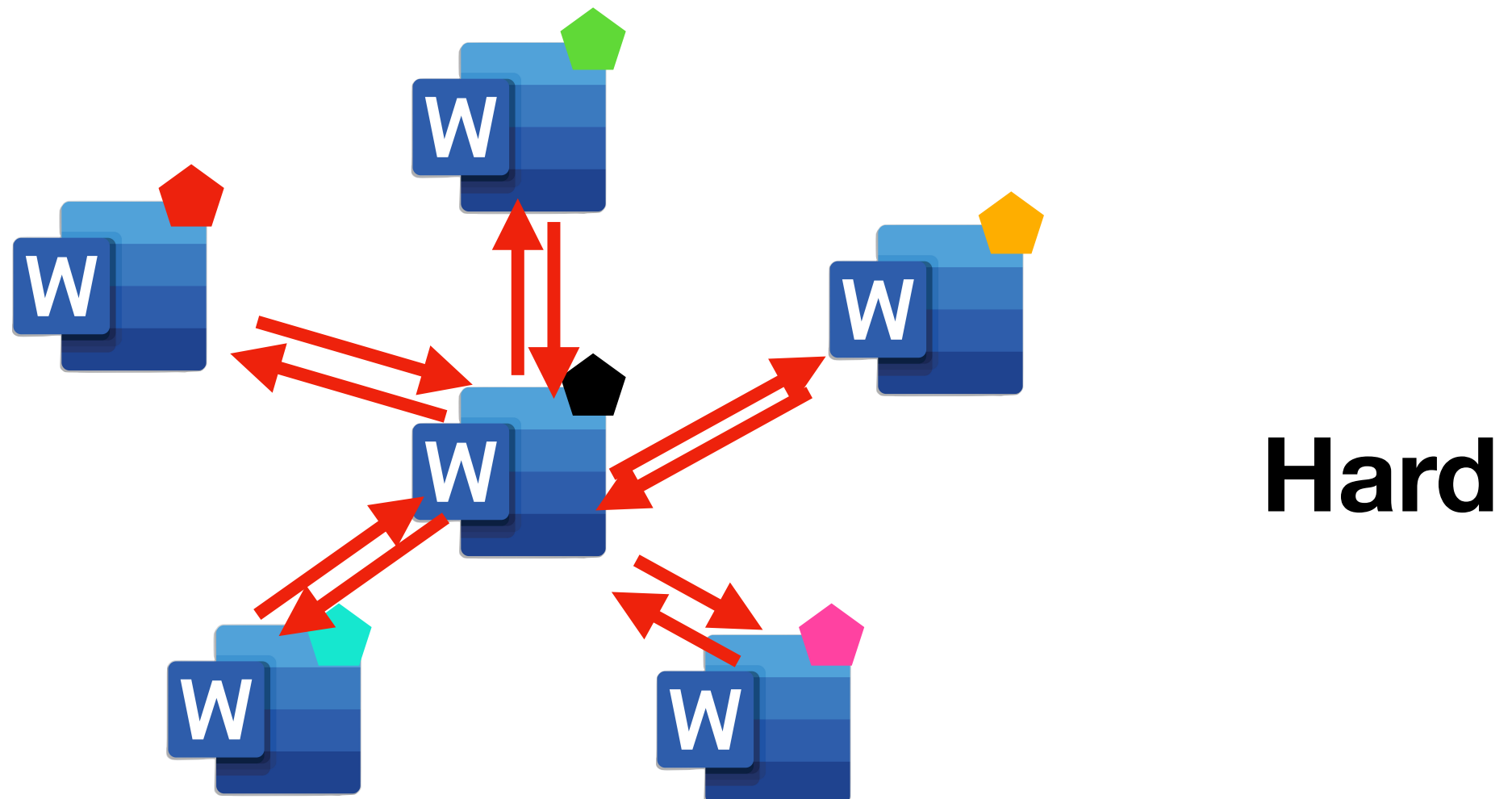
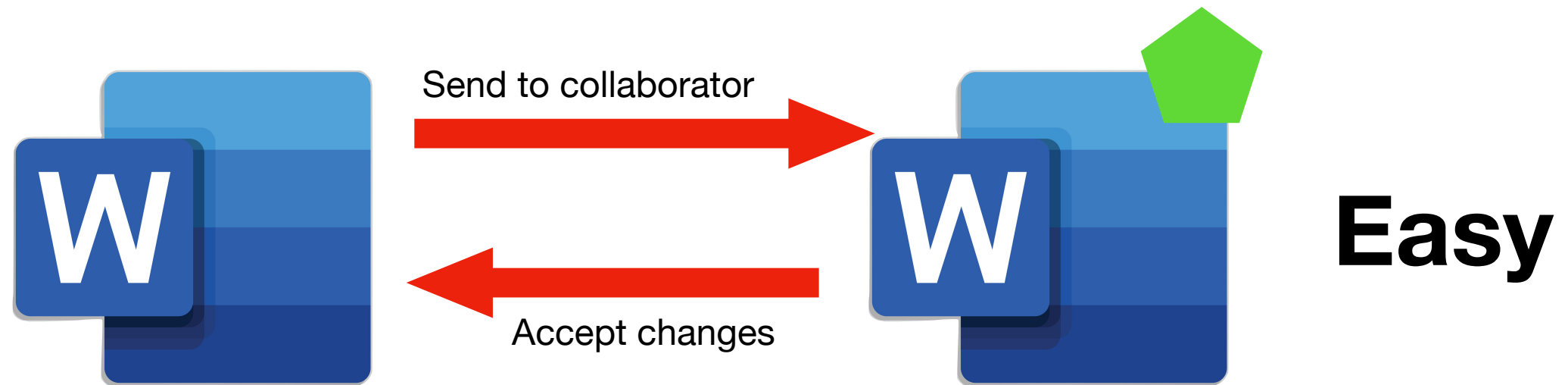
Git

solves two important problems

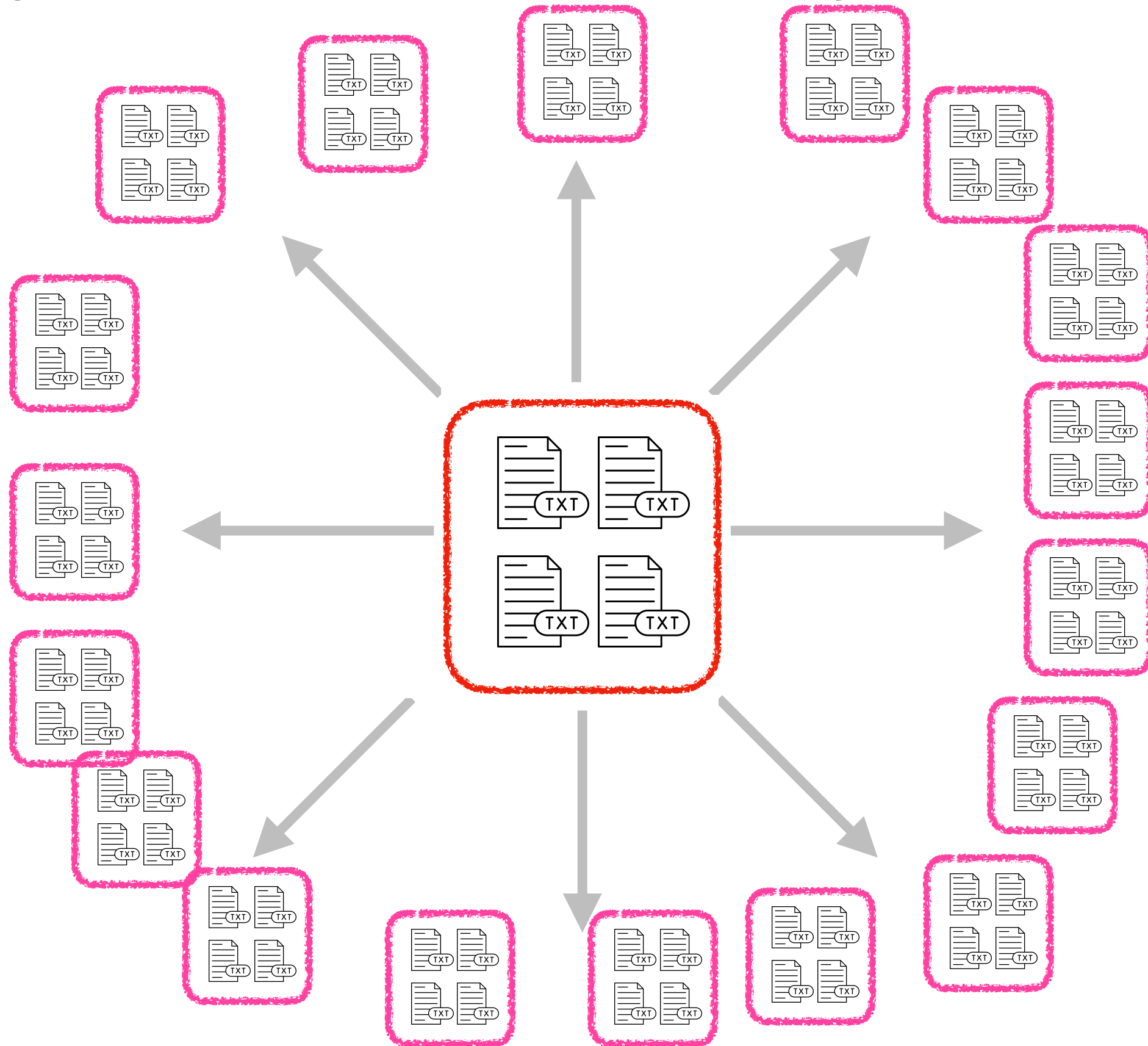
1. Keeping track of changes in plain text documents



2. Facilitating collaborative editing by many people at once



2. Git makes highly collaborative editing of sets of documents manageable for many individuals working asymmetrically



Git terminology

- Git -- the software tool / system for version control
- Repository / repo --
 - Main repository -- the primary location where documents and the versioning information is stored and shared. Usually accessible over a network / the web.
 - Local repositories -- different copies of the repo where individual users are making edits to documents
- Branches -- alternate / parallel versions of your code base

Github
is a commercial host for
git repositories

Github tour

- Create a private repository for your homework assignments (see class wiki for naming instruction)
- Grant Greg and Paul access to your private repo
- Create a Quarto markdown document on your laptop
- Upload your markdown document to Github
- Make some changes locally to your markdown document
- Re-upload your markdown document to Github
- See versioning and time stamp information on Github
- Create a wiki in your private repository (optional; requires Github Education access)